Paper 2

1	(a)	time = 0.43 / 1.1 = 0.39 s.	A1
	(b)	height = ½ gt^2 = ½ (9.81)(0.39)^2 = 0.75 m displacement = $\sqrt{0.43^2 + 0.75^2}$ = 0.86 m	C1 C1 A1
	(c) (1) (2)	horizontal line at a non-zero value of *a*. curved line from origin with increasing gradient.	A1 A1
	(d)	Distance is the <u>actual (curved) path</u> followed by the ball and Displacement is the <u>straight line / minimum distance</u> A to B.	B1
	(e)	acceleration (of free fall) is unchanged / not dependent on mass and so no effect on time taken.	B1